

Field Trials of the Integrated BreadBoard 3: Demonstrating a Rover System in the Sample Fetch Rover Mission Context

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Abstract— The Mars Sample Return campaign constitutes a complex, long-term series of missions aiming at returning samples, contained in Returnable Sample Tube Assemblies (RSTAs), from the surface of the red planet to Earth. As part of this campaign, the Sample Fetch Rover (SFR) was conceived as a European contingency to NASA's Perseverance rover. The purpose of SFR was to fetch and deliver these samples to Mars Ascent Vehicle that would later put them into orbit. Although the concept of this rover was dropped in 2022, it drove the development and validation of autonomous traverse strategies along with autonomous sample pickup and storage operations. Useful for future planetary or lunar exploration missions, these applications are the context and motivation behind the field trials of the Field Test Rover System during the Integrated BreadBoard 3 project.

This paper presents the results of the first real-world demonstration of such a robotic system. The IBB3 field trials took place in September 2023 at a silica sand quarry that covered a breadth of test scenarios, including benign, flat surfaces as well as irregular and rocky terrain. This test campaign served to build confidence in the GNC Mobility and RSTA Acquisition System systems and to identify any difficulties early in their development phases. In particular, this campaign encompassed a series of use cases for both systems: autonomous traverses using the novel Absolute Global Localisation algorithms and the sample fetch and pickup with a high level of autonomy.

I. INTRODUCTION

Bringing back samples from Mars is in the spotlight of space agencies around the world. It is of importance to increase our understanding of the Martian surface properties using instruments that cannot be sent to the red planet. In 2020, NASA launched the Perseverance rover [1], the first element in the multi-mission program. The Mars Sample Return (MSR) campaign [2], planned as a collaboration between NASA and ESA, had foreseen the use of a European rover that would fetch some samples left by Perseverance in a depot. Unfortunately, this rover concept did not progress, but triggered the development of rover technologies that could prove useful in future missions. In particular, autonomy solutions to drive long distances with relatively low drift and manipulation skills to grab and store objects of interest.

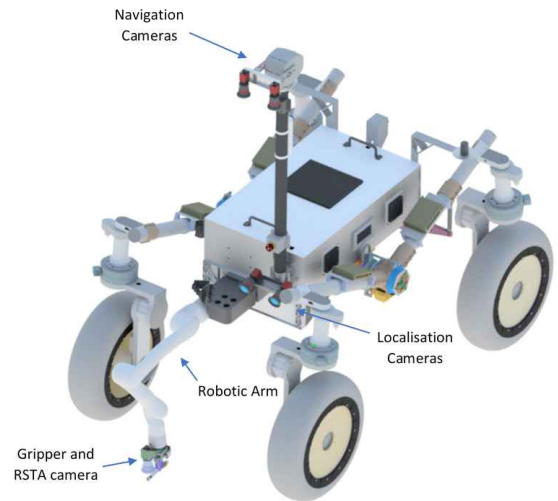


Figure 1. Overview of the FTRS sensors and equipment

Airbus Defence and Space, contracted by ESA, carried out activities towards the development, verification and validation of breadboard technologies in the frame of **Integrated BreadBoard 3 (IBB3)**. The development of the IBB3 activity lasted about 9 months, and took advantage of work on previous breadboard activities. The main novelty for the rover used, the Field Test Rover System (FTRS), was the addition of a robotic manipulator arm on its front, as depicted in Fig. 1. FTRS was tested during the IBB3 field trials of September 2023 inside a sand silica quarry located in Leighton Buzzard, UK (see Fig. 2), the same location as its predecessor activity, IBB2 [3].



Figure 2. Bird's eye view of the quarry test area (L.B. Silica Sands)

This paper presents the results of the trials, focusing on the behaviour and capability of the full end-to-end system.

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First, Section II gives an overview on the different novel autonomy technologies that were tested, including autonomous traverse modes and sample tube retrieval with the robotic arm. Section III presents the results of each of the use cases planned and executed during the trials. Thereafter, Section IV introduces the lessons learnt from the tests, while Section V ends with a series of conclusions.

II. SYSTEM OVERVIEW

IBB3 focuses on three different systems. First, the GNC Mobility system uses four driving modes to traverse scenarios using different grades of autonomy: FollowPath, CheckPath, FOPSA and AutoNav [4] [5]. Second, the **RSTA Acquisition System (RAS)** is used to pick up, capture and store **Returnable Sample Tube Assemblies (RSTAs)** on-board. Third, the Mission Management Layer (MML). The first two are installed on the same mobile platform as in IBB2 [3]. All of them are described thereafter.

A. Traverse Modes

FollowPath (**FPath**) is the GNC mode with the lowest autonomy level. In this mode, the rover performs body level closed loop control to follow a predefined path or to perform a point turn at a specified location. In this mode, the rover follows the path without checking for any obstacles or hazards on its way, leaving the task of ensuring its safety entirely to ground control.

CheckPath mode operates in a similar way to FPath mode. The main difference being that the rover captures images with the navigation cameras at regular intervals and processes them with the Perception and Navigation algorithms. If a hazard is detected, the rover will abort the execution of the path. Therefore, this mode allows the ground operator to specify a path knowing that the rover will autonomously check the safety of this path as it follows it.

FOPSA mode (Follow Path with Safety Assessment) is designed for driving over terrain that has been determined from orbital imagery analysis to be benign, with minimal slope, low slip, and low Cumulative Fractional Area (CFA). The premise of this mode is that the drive performed by following a long term path, commanded by ground control directly, is more efficient than when using AutoNav. The main reasons for this being the reduced number of perception operations whenever the rover stops, as well as a longer distance between stops. This long term path defines the desired route from the starting point (e.g. the Sample Retrieval Lander (SRL)) to the destination (e.g. the Sample Depot) via a series of waypoints. This mode attempts to drive around any detected obstacle on its way if easily avoided, or aborts the path otherwise.

AutoNav mode (Autonomous Navigation) presents the highest level of autonomous operation available on the rover. It is designed for driving over terrain where there is a high probability of obstacles, which are either not avoidable, or not observable in orbital data. The premise of this mode is that the terrain is complex and needs to be assessed for hazards, more thoroughly than with FOPSA mode, to

determine a safe route through the obstacles and achieve the final traverse goal.

For the latter two modes, a new feature was developed in IBB3. It is the ability for the operator to define a Navigation Corridor: a margin around the segment connecting two waypoints with a specified width. This could be used to prevent these higher autonomy modes from leaving the enclosed region, which the operator has deemed to be safe & benign (meaning efficient for traverse) based on orbital imagery, while traversing between waypoints.

Furthermore, all the above-mentioned modes are complemented by Relative Localisation (RelLoc) and Trajectory Control (TrajCtrl) subsystems. RelLoc is based largely on Visual Odometry (VO). However, the need to arrive at a fixed position on the Martian surface (i.e. the Sample Depots) with the high accuracy required by the SFR mission exceeds the capability of RelLoc alone. To overcome this problem, a novel on-board **Absolute Global Localisation (AGL)** algorithm is needed. AGL compares rover images with reference satellite image data stored on-board, allowing the localisation of the rover position with respect to those reference images [6].

B. RSTA Acquisition System (RAS)

Once the rover has arrived at a precise location, close to an RSTA, the RAS is activated to execute the pickup sequence. This system comprises all the functions and equipment necessary to perform the end-to-end sequence of operations including:

- RSTA identification in front of the rover
- Planning of gripper poses to grasp the RSTA
- Deployment of the manipulator and its gripper to grasp the body of the RSTA
- Placement of the RSTA inside the **RSTA Re-grip Bracket (RRB)** to later grasp it again by its head instead of its body
- Insertion of the RSTA into a slot in the **RSTA Storage Assembly (RSA)**

The RAS consists of a suite of equipment, software applications and hardware including [7]:

- Acquisition Management Software (AMS)
- Vision Based Detection Subsystem (VBDS)
- Arm and Gripper Subsystem (AGS)

C. Mission Management Layer (MML)

The commanding of all Mobility and RAS operations were done through an operator visualisation interface (3DROCS) and routed through the Functional Task Manager (FTM), which is a Mission Management Software (MMS) emulator developed for the project.

The FTM contained all the logic to carry out complex multi-step test sequences autonomously with no operator input, correctly flagging any unexpected events and responding appropriately, continuing operations without ground input if possible. The objective of this addition is to provide accurate simulation of the SFR mission scenario

where transmission delay and limited communications passes/bandwidth makes live operator input impossible.

3DROCS provides a view of the rover on the terrain derived from rover telemetry, as well as the ability to schedule an activity plan of complex tasks to be executed by the FTM.

III. IBB3 FIELD TESTING RESULTS

A series of use cases were conceived for the IBB3 field trials to validate the functionalities presented in the previous section. The first three tests focus on the autonomous traversal capabilities of the system, another three on the pickup of sample tubes and the last one on a newly developed capability of heading correction. The tests were executed in September 2023 and their results are presented next.

A. Long Range Traverse - Nominal Use Case

For the Long Range Traverse use case the rover was commanded to perform a long range traverse to reach 5 waypoints using both FOPSA and AutoNav modes with a max rover speed of 8.25 cm/s. AGL was also used during this test to maintain the rover position and attitude, as well as correcting for errors it was intentionally initialised with, 5 metres and 1 degree respectively. Over the test duration, the rover autonomously covered a distance of over 150 m.



Figure 3. FTRS performing a Long Range Traverse at field trials in a sand quarry in the UK

The objective of the test was the demonstration of the AutoNav and FOPSA functionalities in an outdoor environment. In particular, the commanded traverse included the operations described in Table I and summarised hereafter:

- Traverse to Waypoint (WP) #1 in FOPSA mode
- Traverse to WP#2 in FOPSA mode
- Traverse to WP#3 in AutoNav mode
- Traverse to WP#4 in FOPSA mode, with an expected automatic transition to AutoNav due to the placement of large obstacle (see Fig. 3)
- Traverse to WP#5 in FOPSA mode

The RelLoc position estimation error after driving 200m, which was clearly beyond expectations, is around 2.5%, which is within SFR flight requirements (interpolated from the 100m and 300m SFR localisation requirements, being 2.3% & 3.4% respectively). TrajCtrl performed well

throughout the test, keeping the control errors low with no replan events (i.e. a failure to hold to the commanded path) - see Table II.

TABLE I. LONG RANGE TRAVERSE - LIST OF WAYPOINTS

Waypoints	Coordinates [m]		Cumulative distance [m]	Cumulative traverse time [hh:mm:ss]	Traverse Mode
	X_{RM}	Y_{RM}			
Start pos.	23.47	4.09	-	-	-
#1	44.48	-56.84	64.45	00:39:32	FOPSA
#2	63.88	-77	92.43	00:56:35	FOPSA
#3	58.96	-93.56	109.70	01:10:11	AutoNav
#4	63.76	-106.12	123.15	01:25:48	FOPSA
#5	83.96	-126.52	151.86	01:42:53	FOPSA

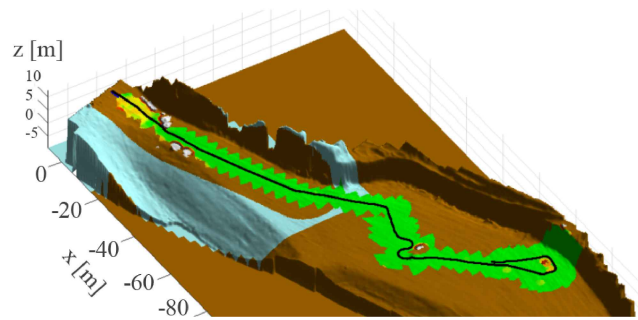


Figure 4. Path and NavMaps on the 3D terrain - Long Range Traverse

In this test, AGL successfully corrected the 5 m of error it was initialised with and correctly maintained a fix of the rover's position throughout the remainder of the test.

TABLE II. PATH FOLLOWING METRICS - LONG RANGE TRAVERSE

Total distance traversed (incl. further WPs)	191.67 [m]
Total accumulated heading change	1798.2 [deg]
Number of Steer/Drive commands	12768
Trajectory Control lateral error	0.092 [m] (max) 0.062 [m] (3 σ)
Trajectory Control heading error	10.0 [deg] (max) 3.4 [deg] (3 σ)
Number of replan events	0

This long traverse, far exceeding the distances of such a traverse which can be driven in Airbus' Mars Yard [8], successfully demonstrated the functionalities required for the sample fetch system to traverse from the lander to the depot site, constraining VO drift by use of AGL without any need for ground intervention.

Compared to the trials of 2022 (IBB2) [3], in IBB3 this Use Case incorporated RelLoc working in closed loop (true for all IBB3 tests), the capability to autonomously transition from FOPSA to AutoNav, Navigation Corridors and the significant integration into an end-to-end system controlled by the operator through 3DROCS.

B. Human Directed Drive - Nominal Use Case

The objective of this test was to demonstrate the robustness of the path planning function in scenarios where it is not possible to plan a navigable path due to the characteristics of the terrain. Fig. 5 shows a sample of the density of obstacles placed. In this scenario, the rover performed the following sequence of actions:

- Traverse to the waypoint indicated by the operator
- The first ~5m of the path are navigable
- Realise that the next part of the path is blocked
- Try to find an alternative path
- Discover the waypoint is, in fact, unreachable
- Abort the operation and wait for the next command

The system will continue to search for alternative paths until it decides there are no routes to the target, or the sol traverse limits are reached (mission level drive time limits).



Figure 5. Human Directed Drive test area

The rover was capable of continuously constructing and updating a Navigation Map (NavMap) while the pre-defined paths from the operators were followed using CheckPath. Fig. 6 shows the NavMap projected onto the 3D reconstruction of the test area.

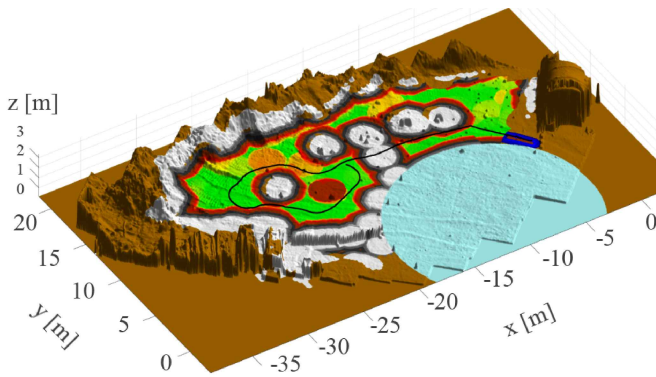


Figure 6. Path and NavMaps on the 3D terrain - HDD use case (cyan - autonomy not allowed zone; red - high risk zone, orange - medium risk zone, green - low risk zone; white - forbidden area)

C. RSTA depot fetching test - Nominal Use Case

The objective of the use case was the demonstration of the autonomous fetching of three RSTAs in a depot in the nominal case. The nominal case consists of the rover succeeding in picking up each of the 3 RSTAs at their

primary **RSTA Pickup Access Point (RPAP)** without operator intervention. In other words, this use case served to demonstrate the capability to autonomously:

- drive to 3 RSTA locations with absolute accuracy of 19 cm (at the pickup location), aided by AGL
- perform sample tube detection
- carry out pickup and storage sequence.

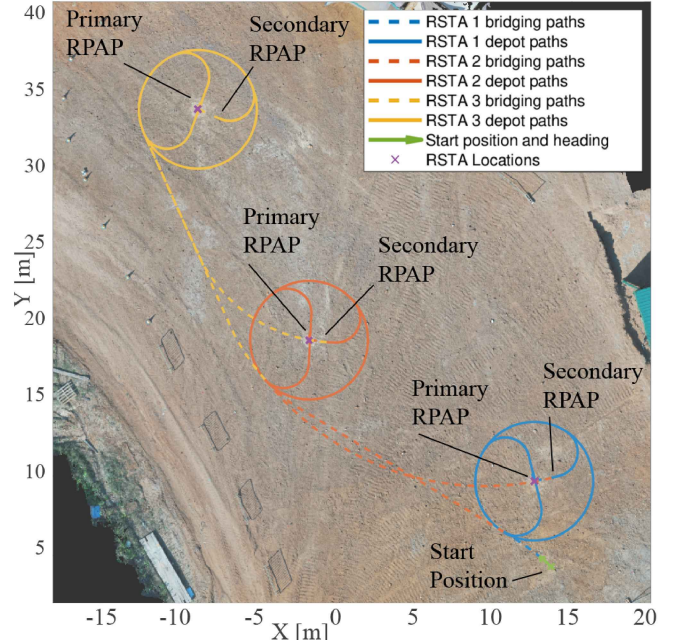


Figure 7. All possible depot paths for all three RSTA locations

Fig. 7 shows the paths created to reach the locations of three RSTAs. For each location, a ring-shaped depot path is created with two RPAPs. The second RPAP serves as a backup in case the first fetching attempt fails. If this happens, the rover drives backwards following the approach path until getting back to the outer ring, so it does not drive over the RSTA.



Figure 8. RSTA depot fetching operations after arriving at the RPAP with visible rover tracks

The RAS was commanded to pick up an RSTA once the Mobility system had successfully arrived at each RPAP (see Fig. 8). It was expected that an RSTA would be successfully picked up (see Fig. 9) and stored at each location.



Figure 9. The FTRS performing sample retrieval operations

During execution of the test the Mobility system successfully followed pre-defined paths to each RPAP (see Fig. 10) without operator intervention, with AGL correcting deliberate initialisation errors (i.e. 10 cm position error, 1 degree heading error). It performed so well that FTRS repeatedly drove in its own tracks.

For all three approaches the system placed the rover such that the RSTA was well within 19 cm of the centre of the RAS workspace (see Fig. 12), the maximum error allocated to Mobility in the error budget.

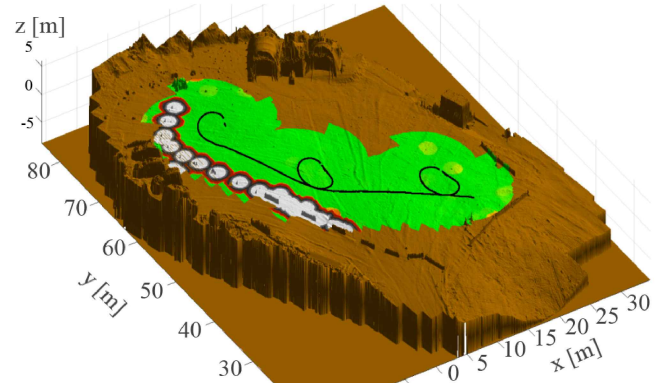


Figure 10. Path and NavMaps on the 3D terrain - Depot use case (green - safe zone; white - forbidden area; black ovals - RPAP locations, with the first one placed at the right side of the image)

The pickup sequence of the RAS, shown and described in Fig. 11, was executed successfully without operator intervention for each RSTA and the test concluded with all three stored in the RSA.

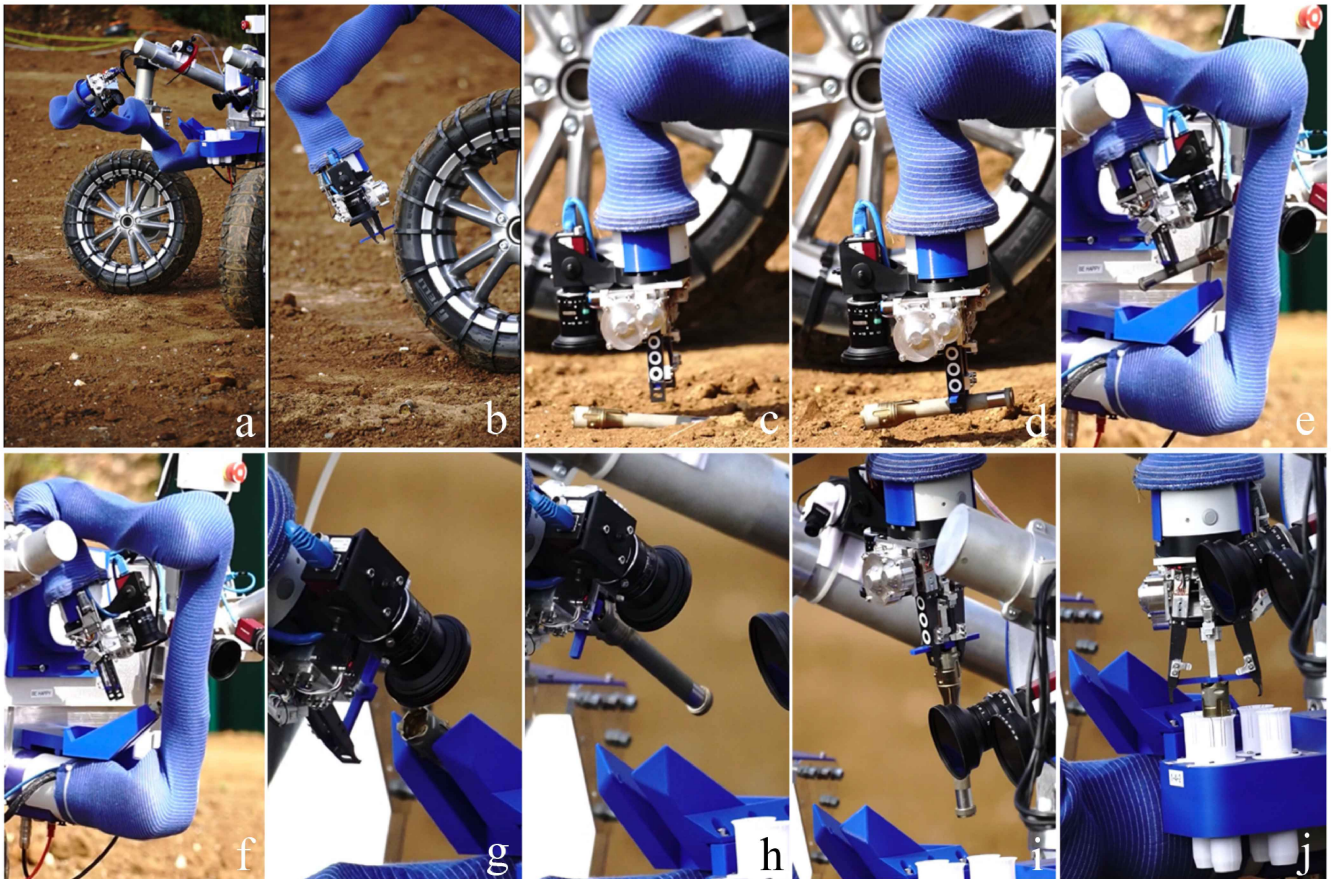


Figure 11. Sequential fetching operation (starting top-left): a) Deployment b) RDC Image Capture c) Descend Grasp Trajectory d) Grasp RSTA e) RSTA Regrip Bracket (RRB) Insertion f) Open Gripper g) RSTA Pickup from RRB h) RSTA Extraction from RRB i) RSA Approach j) RSA Slot Insertion

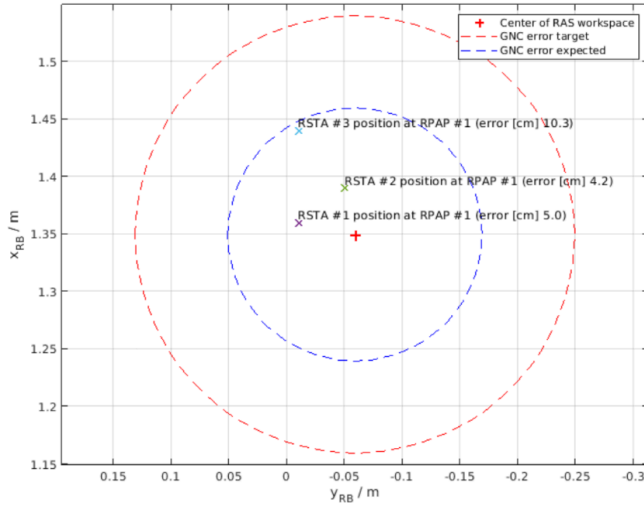


Figure 12. Three RSTA positions at RPAP with the error budgets during nominal case

D. RSTA Human Directed Pickup Test - Nominal

This test was to demonstrate the capability to pick up RSTAs with pre-planned Human Directed Pickup (HDP), mixing human directed and autonomous operations. This operational mode is a contingency for cases where ground control assesses in advance the scenario and determines that fully autonomous acquisition is not feasible or too risky due to the geometric arrangement.

The objective was to demonstrate the ability of FTRS to perform a hybrid autonomous-directed fetch operation following these steps:

- Autonomously drive pre-planned path to RPAP.
- Point NavCam towards the RSTA workspace and run NavCam VBD to provide ground with images and point cloud for assessment.
- Estimation of grasp pose by the operators with ground HDP tool using the aforementioned images and point cloud. Prepare and execute an activity plan with the estimated grasp pose to pick up the RSTA.

The depot paths are the same as in the previous use case, but with only the first RSTA. Once the Mobility system had successfully arrived at the RPAP, a human operator injected the pose of the RSTA directly after checking the image data captured by the rover (see Fig. 13). On arrival at the first RPAP, the RSTA was found to be on the left side of the RAS workspace (see Fig. 14). Manually stereo correlating the position of the RSTA centre at the primary RPAP in post-processing shows it to be 16 cm to the left of the workspace centre; this agrees with the calculated AGL error relative to ground truth, which was 17 cm in X direction.

This test partially met its objectives. The rover autonomously drove a pre-planned path to the first RPAP, placing itself with respect to the RSTA within the GNC error limits. Thereafter the NavCam provided imagery and point cloud of the RSTA workspace, which helped ground operators to estimate the grasp pose. After four unsuccessful attempts to correct the grasp pose, changing the command

input specifying the orientation in a different quaternion order on the fifth attempt resulted in the correct pose to grasp the RSTA. However, the execution of the activity plan failed when the arm could not retrieve the RSTA from the RRB (RSTA Regrip Bracket), due to the arm moving before the gripper was fully closed.

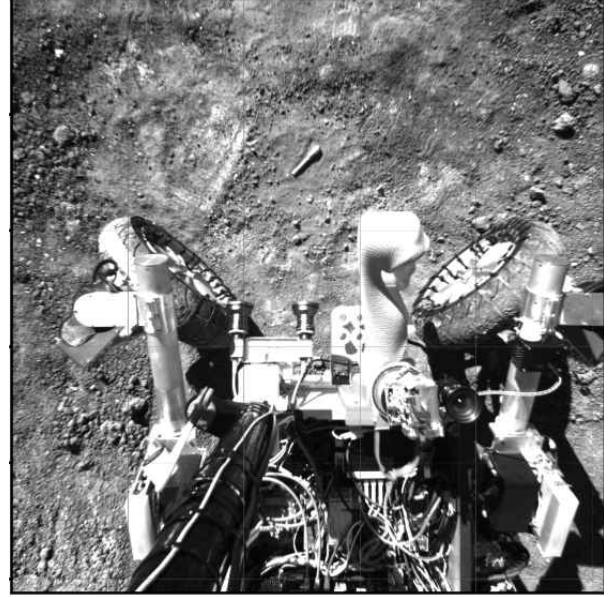


Figure 13. NavCam image acquired at the arrival of the third RPAP

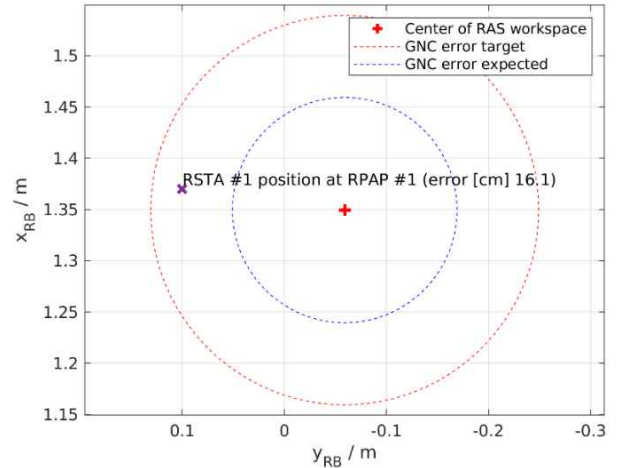


Figure 14. RSTA position at RPAP with error budgets during corner case

E. Sun Sensing Heading Estimation - Correction Case

The purpose of this particular use case was to demonstrate the capability to correct a heading error with respect to a planet-fixed coordinate system by performing autonomous **Sun-Sensing Heading Estimation (SSHE)**. This algorithm generates a full rover attitude estimation with respect to a global frame from an image of the sun and accelerometer measurements. The image of the sun is used to determine the sun vector relative to the rover, and therefore the rover heading given a-priori ephemeris data, knowledge of the time the image was taken and the rough rover position; combined with the gravity vector determined

from accelerometer measurements this provides the full rover attitude.

This stress test consisted of using SSHE to reliably correct the rover heading while performing an autonomous traverse. Hence, the rover was initialised with a substantial initial heading error of 2.5 degrees and commanded to a first waypoint at a distance of about 20 metres. After arriving at the respective waypoint, the sun observation and heading correction was performed, as denoted in Fig. 15. The rover then drove for another 20 metres in a straight line to show the effect of the corrected heading.

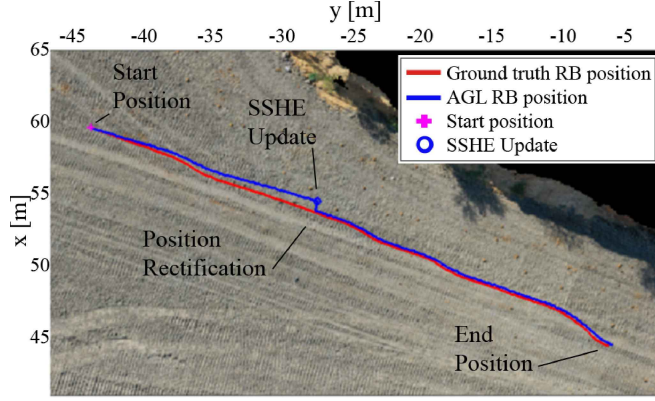


Figure 15. AGL and ground truth position with path heading correction

As shown the sun detection and localisation algorithm was able to:

- identify the sun shape as the only valid contour
- identify the correct outline and centre of the sun (contour) in the image plane
- generate the pixel coordinates of the sun centre

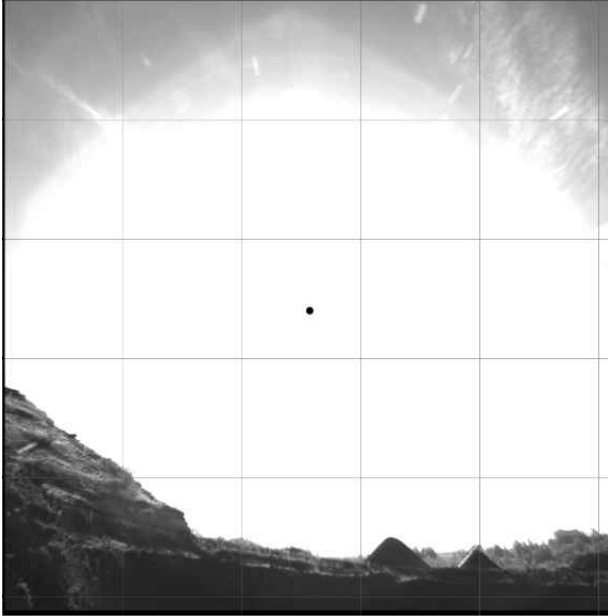


Figure 16. Sun detection NavCam image captured autonomously by SSHE
Note: the sensors used by the rover cameras exhibit the “black sun” effect

Furthermore, the autonomously commanded Pan and Tilt Unit (PTU) orientation managed to capture the sun within the FoV of the cameras (see Fig. 16). PTU angle commands

were autonomously generated by the SSHE based on the estimated direction of the sun, given the known rover position and absolute on-board time. The heading measurement generated by SSHE was correct within the precision of the ground truth system deployed for the field trials, completely accounting for the initial heading error. Therefore, the behaviour of the sun detection and localisation function of SSHE surpassed initial expectations.

Additionally, AGL possesses an interface to accept heading corrections which was exercised during this test using the output from SSHE. Fig. 17 shows the AGL heading estimation error, relative to ground truth. For this test, AGL was initialised with an initial heading error of -2 degrees, while the AGL reference map was modified such that no valid AGL could be achieved during this test. This shows that the only heading correction carried out is the one injected from the sun sensing measurement (the jump of signals clearly visible at 17.5 metres) and successfully demonstrates the concept envisioned for the SFR mission of augmenting heading knowledge using SSHE measurements.

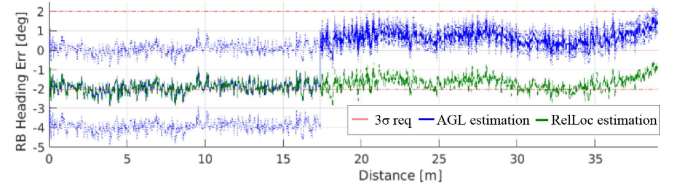


Figure 17. Heading estimation over driven distance in the case of SSHE correction

IV. LESSONS LEARNT

Two weeks after the end of the field trials, the IBB3 team gathered all the information that could improve the experience in future test campaigns. The feedback and comments collected were not limited to any particular topic or discipline, and covered not only the field trials period but also the preparation phase. Positive feedback and opportunities were also raised when considered important. A range of topics have been covered such as team organisation, test setup, test procedure, training before the trials, and specific discipline-oriented work.

The preparation phase covered a period between May and beginning of September 2023. It effectively started when the robot arm used by RAS was integrated on FTRS. It was also when the different IBB3 teams (Mobility, RAS, operations, software, platform) were required to work more closely, coordinating the sharing of rover time and resources.

The main lessons learned from this phase is that starting shakedown in the Airbus car park and Mars Yard two months before the trials was well worth it. The authors concluded that:

- it reinforced the link between the different teams taking part in the testing
- it gave time to the analysts and the operators to get used to the operations environment (e.g. use of procedures and 3DROCS) and the test environment (e.g. test the ground truth system, set up of the Starlink internet solution, use the recording system)

- it gave more knowledge of the entire system to the team, and gave more confidence to be efficient during the field trials

Nevertheless, we considered that some elements of the system were not tested enough during the shakedowns. This resulted in more pressure, and less effective time for formal testing during the trials at the quarry.

V. CHALLENGES

Challenges encountered were focused on system robustness, both at the integrated system level, when the network load was intense, and at the subsystem level. In particular the RAS suffered from a lack of robustness - for which lessons learnt have been captured and the resolutions planned, to be implemented before further testing and IBB4. Insufficient preparation and integration testing was undertaken ahead of trials due to a challenging timeline and limitations of facilities for representative testing. There was also insufficient maturity of 3DROCS integration. Despite continuous co-engineering and successful colocation, it is clear that more time and collocations would have been needed to bring the operations interface to full maturity. Finally, the VBDS was not robust to the variety of terrain and lighting conditions in a field trials situation, compared with previous testing in more controlled conditions.

VI. CONCLUSION

The IBB3 field trials achieved a considerable step forward, from IBB2, in taking a fully integrated end-to-end system including an MMS emulator and the operator commanding interfacing (3DROCS [9]) to field trials. The operational concept and key technological functionalities to retrieve samples from a depot were demonstrated successfully with the addition of the RAS and several entirely new Mobility functionalities were successfully showcased (as described in Section II). Considerable experience has been gained, lessons learnt and some clear objectives for improvement identified. Overall, the trials are viewed as being a great success, with scope for further improvement particularly in regards to overall system robustness.

Partially successful testing of a fully integrated system including MMS emulator and operator command interface. A number of maturity issues, in particular with 3DROCS and its interfacing with the on-board ROS2 [10] network, prevented in some test cases continuous operations and required the use of workarounds. Nonetheless, it was possible to demonstrate the validity of the design and concept of operations. Future work will be required to fix implementation and robustness issues and improve system operability.

The combination of TrajCtrl, RelLoc and AGL resulted in a highly performant driving system in the depot areas. The rover was observed to drive in its own tracks for subsequent tests and repeatedly arrived at the RPAP with the accuracy required to support RAS operations. It was also demonstrated that the localisation capability and performance outside of the depot could be even further

augmented with the addition of SSHE.

IBB3 delivered significant advancements in field learning, both at the overall system level and within the various components of the system. The trials also facilitated successful demonstrations of new Mobility functionalities which could be ported back to existing missions as well as being deployed in future missions (SSHE - presented above and 'Efficient Navigation' - outside the scope of this paper). Successful demonstration of the end-to-end concept of operations and partially successful demonstration of functionality for autonomous fetching of RSTAs in the depot, including precision placement of the rover using the new functionality AGL developed for SFR and first demonstrated in the field at IBB2, Visual Based Detection of the RSTA (GMV), autonomous grasping pose determination, arm trajectory planning and execution, and the sequence of manipulations required to collect a sample tube from the surface and re-grip such that it can be stowed on board.

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